Surv 617 HW1

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Overview of data

In a subset of data from the Health and Retirement Study, 311 participants from the "War Babies" cohort (born 1942-1947) had their systolic blood pressure checked every 4 years. This subset was taken from the RAND HRS Longitudinal File 2022. The dataset contains the following variables:
-HHIDPN: Six character person ID, combining HRS Household ID and HRS person number
-bp_sys: Systolic blood pressure (mmHg)
-wave: HRS wave
-age_y: Age in years at a given wave
-year: Numeric measure of years since first blood pressure measurement
-RAGENDER: 1 = male, 2 = female

Study objectives

1. On average, how does systolic blood pressure change over time as people get older?

2. How much variance among participants is there in the trajectories of systolic blood pressure, and can this variance be explained by sex and age?

Questions

Complete an analysis as described below to address the study objectives using your judgement in order to keep things as simple as possible.

1.

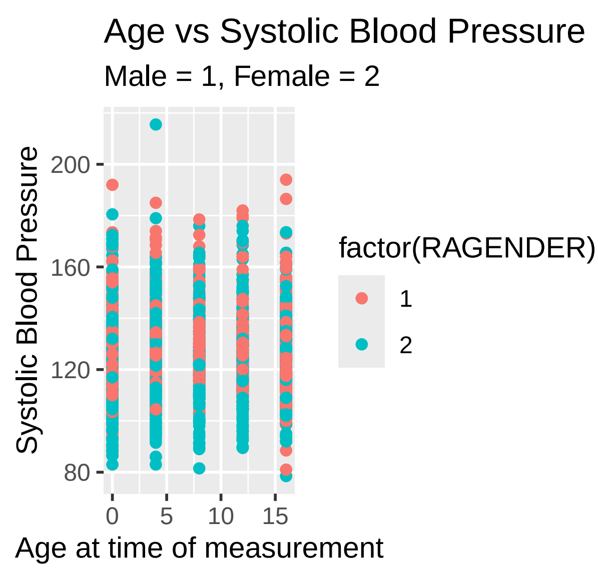
Describe the sample in a reasonable way, i.e., construct a table or a graph of descriptive statistics as you may do for a publication. Be sure to remember that there is more than one observation per subject. Comment briefly on the general characteristics of the observations. Keep the study objectives in mind with your descriptive presentation.

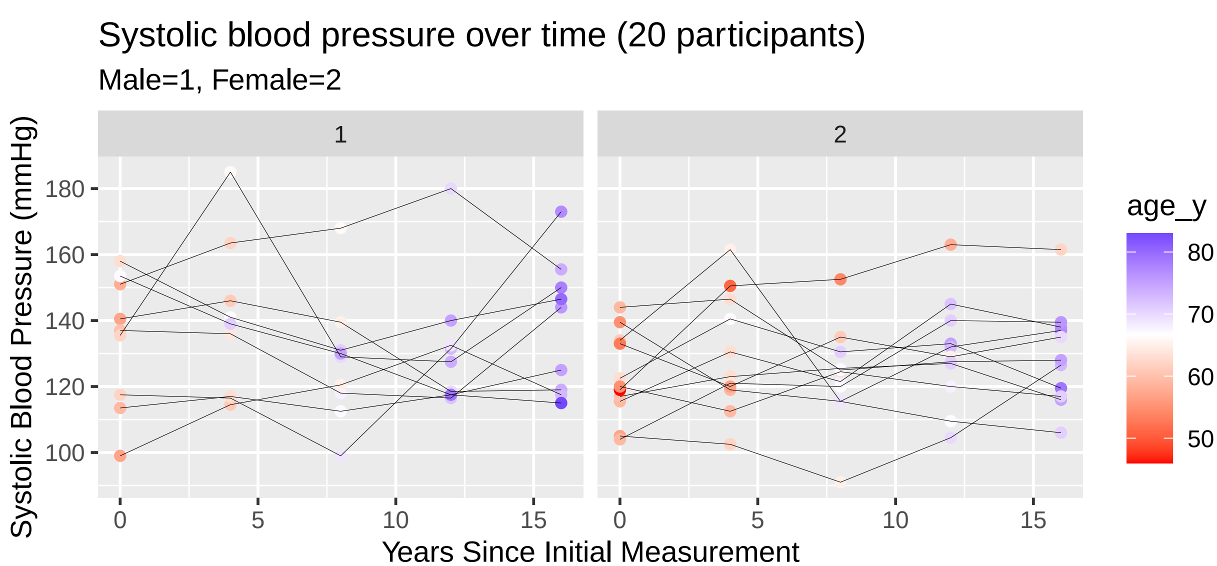
```
## [1] "Participants: 311"
```

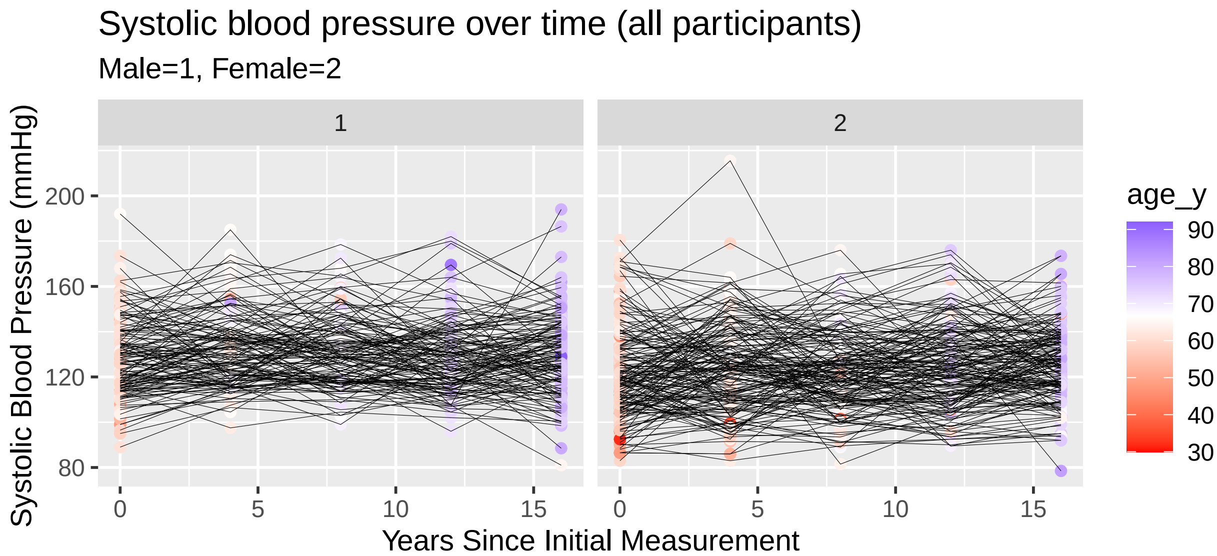
```
##       bp_sys        RAGENDER
##  Min.   : 78.5    1:685
##  1st Qu.:114.5    2:870
##  Median :125.0
##  Mean   :126.5
##  3rd Qu.:137.5
##  Max.   :215.5

## [1] "Starting Age Summary"

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   30.00   58.00   59.00   58.77   62.00   76.00
```







The data includes 311 participants, each with five observations - one initial measurement, and follow-ups every four years over the course of 16 years. 55.9% of the sample were female, 44.1% male. The age of participants when they entered the study ranged from 30 to 76 years, with an average of 58.77. According to the box plots, the age of males in the study is generally higher than females, and women have far more statistical outliers on the lower tail of the age distribution.

The second plot places systolic blood pressure against time, which relates to the first objective of the study. It is difficult to determine the average change in blood pressure over time, as the range of values at the beginning of the study is very similar to the end. The final plot reveals more clearly that many participants have a higher systolic blood pressure at the end of the study than at the beginning, corroborating the previous point. The third plot also shows the systolic blood pressure over time for the first 20 (reduced for simplicity/readability) participants, 9 males and 11 females. The left half shows males, the right shows females, and there is a color encoding to represent the age of participants. This plot could be useful in exploring the second objective of the study, as it shows the relationship between blood pressure and time, along with encoding sex and age. Age and years since initial measurement are inherently correlated, so many of the points to the right side of the plot are blue (older participants). We see again that women in the study are generally younger than the men, with "redder" points in general. There appears to be a great deal of variance in the trends of blood pressure over time. The *effect* of age is less clear, and could requires alternative visualizations in addition to the forthcoming statistical analyses. There is no obvious pattern with trend lines and starting age, nor with starting points necessarily. However, it is quite clear that each subject has a unique trend-line and intercept, which strongly motivates the use of hierarchical modeling.

2.

Fit a multilevel model with systolic blood pressure as the dependent variable and time (year) as the independent variable to explore how systolic blood pressure may have changed over time. Make sure to account for the within-subject correlation in observations on the dependent variable in the multilevel model that you are fitting. Be sure to clearly describe your reasoning.

```
# subjects at level 2, repeated measurements over time at level 1
# (1|HHIDPN) only allows intercept to change
# (year|HHIDPN) allows slopes to change as well
```

```r
model1 <- lmer(bp_sys ~ year + (year|HHIDPN),REML=T,data=hrs_wbbp_long)
summary(model1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: bp_sys ~ year + (year | HHIDPN)
##    Data: hrs_wbbp_long
##
## REML criterion at convergence: 12974.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.2159 -0.5805 -0.0450  0.5128  4.1369
##
## Random effects:
##  Groups   Name        Variance Std.Dev. Corr
##  HHIDPN   (Intercept) 203.7644 14.2746
##           year          0.5133  0.7165  -0.58
##  Residual             162.3796 12.7428
## Number of obs: 1555, groups:  HHIDPN, 311
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept) 125.5794     0.9841 309.9737  127.61   <2e-16 ***
## year          0.1164     0.0701 309.9867    1.66   0.0979 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## year -0.654
```

The reasoning behind the above model was to establish a simple model that could capture the average relationship between time and blood pressure (objective 1), and to allow this relationship to vary among subjects. We saw from the third visualization that patients have a wide range of trajectories, thus allowing the slopes and intercepts of blood pressure over time to vary is logical.

3.

Write out the model that you fit in question 2 using mathematical notation.

$$BP_{ij} = \beta_0 + \beta_1 YEAR_{ij} + \gamma_{0j} + \gamma_{1j} YEAR_{ij} + e_{ij}$$

Where the subscript ij represents the ith measurement of the jth participant in the study. β_0, β_1 are the fixed effects (average blood pressure at the start of the study, and overall effect of year on blood pressure, respectively), and γ_{0j}, γ_{1j} are random effects (for the intercept and slope of the jth participant, respectively) distributed by

$$\begin{bmatrix} \gamma_{0j} \\ \gamma_{1j} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_0^2 & \sigma_{01} \\ \sigma_{01} & \sigma_1^2 \end{bmatrix} \right)$$

and $e_{ij} \sim N(0, \sigma_\epsilon^2)$

4.

Interpret the model parameters estimated in part 2, including any estimated variance components. Frame your interpretations with respect to addressing the two research objectives.

The model estimated the six parameters, with interpretations as follows:

- $\hat{\beta_0} = 125.57$. Fixed effect suggesting the overall average systolic blood pressure at the start of the study is around 125 mmHg. This is the expected starting blood pressure if all other predictors are 0, which simply represents the start of the study for this model.
- $\hat{\beta_1} = 0.1164$. Every year passed results in an increase in systolic blood pressure of 0.1164 mmHg, averaged across all participants and controlling for within-subject variance. This is an answer to the first objective - systolic blood pressure increases over time on average. In other words, this is an estimate of the *longitudinal* effect of time on blood pressure.
- $\hat{\sigma_0^2} = 203.7644$. The variance between intercepts among subjects seems to be quite high, meaning the starting blood pressure of participants is widely dispersed. We saw this dispersion in the boxplots of the exploratory data analysis. This reaffirms the value of a multilevel model to capture this variability.
- $\hat{\sigma_1^2} = 0.5133$. The variance between slopes among subjects appears comparatively small. This gets at an answer to the second objective - there appears to be variance between the blood pressure trajectories of participants, though tests of significance are required to give this weight (see next section). It should be stated that this variance captures the variability in the *cross-sectional* effect of time on blood pressure.
- $\hat{\sigma_{01}} = -5.295$. A negative covariance/correlation between random slopes and intercepts means that lower initial systolic blood pressures (intercepts) correspond to larger increases of blood pressure over time (slopes). In other words, the slopes vary such that blood pressure at the end of the study is more closely grouped together than at the beginning. In terms of this problem, patients that start with high blood pressure may generally be expected to stay around the same level as time goes on, whereas patients starting with low blood pressure may generally be expected to experience a larger increase in their blood pressure over time. More analysis would be needed to determine the statistical significance of this parameter.
- $\sigma_\epsilon^2 = 162.3796$. There is still a fair amount of variability remaining in the errors, which likely means that some of the data has been fit well but other data has been fit very poorly, leading to high variability in the errors. This implies (in addition to other factors) that other models must be considered.

5.

Briefly describe the steps that one would use to test the null hypothesis that the variance of random year effects is equal to zero, including the computation of the p-value for this test. What is the conclusion of this test as it applies to the model fit in question 2 with respect to the research objectives?

For this test, we must fit a model which *does not include* the random coefficients and compare it to the original model. I will call this "model1a" in the following code. Once this has been done, a likelihood-ratio test can be conducted, with the test statistic calculated as the difference between the -2xlog-likelihoods of the models. This is known to follow a chi-square distribution, but since the models have a different number of random coefficients, we use a chi-square mixture. The following shows the p-value to be around 0.00000266, allowing us to reject the null hypothesis that there is no variance in the random effects of year. In other words, we conclude that there is significant evidence that the effect of year *varies across study participants*, which tells us that the trajectories of blood pressure over time *does* vary, giving insight to the first objective. The next model will incorporate age and sex as potential explanatory factors of the variance in blood pressure trajectories.

```r
lrt_func <- function(fit1,fit2){
  lrtstat <- (-2*as.numeric(summary(fit1)$logLik)) -
    (-2*as.numeric(summary(fit2)$logLik))
  pval <- 0.5 * (1-pchisq(lrtstat,2)) +
    0.5 * (1-pchisq(lrtstat,1)) # 2 random effects compared to 1
  return(pval)
}
model1a <- lmer(bp_sys ~ year + (1|HHIDPN),REML=T,data=hrs_wbbp_long) # random intercepts only

lrt_func(model1a,model1)
```

```
## [1] 2.668431e-06
```

```r
model1b <- lmer(bp_sys ~ year + (year||HHIDPN),data=hrs_wbbp_long)
# also testing whether correlation between random effects should be included
lrt_func(model1b,model1)
```

```
## [1] 1.042782e-05
```

```r
# yes, include the correlation, significant improvement to log-likelihood
```

6.

Modify the model to account for the sex and age effect on systolic blood pressure.

```r
model2 <- lmer(bp_sys ~ year + age_y + RAGENDER + (year|HHIDPN),
               REML=T,data=hrs_wbbp_long)
summary(model2)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: bp_sys ~ year + age_y + RAGENDER + (year | HHIDPN)
##    Data: hrs_wbbp_long
##
## REML criterion at convergence: 12935.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.3015 -0.5781 -0.0526  0.5187  4.2482
##
## Random effects:
##  Groups   Name        Variance Std.Dev. Corr
##  HHIDPN   (Intercept) 170.7487 13.0671
##           year          0.5073  0.7122  -0.54
##  Residual             162.1771 12.7349
## Number of obs: 1555, groups:  HHIDPN, 311
##
## Fixed effects:
##              Estimate Std. Error       df t value Pr(>|t|)
```

```
## (Intercept)  83.5814     9.2466 322.5783   9.039  < 2e-16 ***
## year         -0.6352     0.1674 448.1111  -3.795 0.000168 ***
## age_y         0.7507     0.1519 320.0203   4.942 1.25e-06 ***
## RAGENDER2    -3.7712     1.4827 308.9165  -2.543 0.011466 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) year   age_y
## year       0.873
## age_y     -0.991 -0.909
## RAGENDER2 -0.369 -0.263  0.289
```

```
# no covariance between random effects
model2a <- lmer(bp_sys ~ year + age_y + RAGENDER + (year||HHIDPN),
               REML=T,data=hrs_wbbp_long)
lrt_func(model2a,model2)
```

```
## [1] 0.0001233501
```

```
# include the covariance parameter, significantly improves the fitted log-likelihood
rm(model2a,model1a,model1b)

# can the variance in trajectories BE EXPLAINED by age and sex - interaction terms
# slopes and intercepts interaction model - singular fit, similar AIC
model3 <- lmer(bp_sys ~ year + age_y + RAGENDER + age_y:year + RAGENDER:year +
                 (year|HHIDPN),
               REML = T,data=hrs_wbbp_long)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.0233733 (tol = 0.002, component 1)
```

```
model3 %>% summary()
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: bp_sys ~ year + age_y + RAGENDER + age_y:year + RAGENDER:year +
##     (year | HHIDPN)
##    Data: hrs_wbbp_long
##
## REML criterion at convergence: 12936.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.2234 -0.5867 -0.0497  0.5160  4.2329
##
## Random effects:
##  Groups   Name        Variance Std.Dev. Corr
##  HHIDPN   (Intercept) 169.2586 13.0099
##           year          0.4838  0.6956  -0.54
##  Residual             161.9762 12.7270
## Number of obs: 1555, groups:  HHIDPN, 311
```

```
##
## Fixed effects:
##                  Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)     7.247e+01  1.069e+01  4.292e+02   6.782 3.93e-11 ***
## year            7.068e-01  6.606e-01  1.067e+03   1.070  0.28491
## age_y           9.421e-01  1.731e-01  4.182e+02   5.444 8.90e-08 ***
## RAGENDER2      -5.278e+00  1.929e+00  3.189e+02  -2.736  0.00656 **
## year:age_y     -2.182e-02  9.392e-03  1.127e+03  -2.323  0.02037 *
## year:RAGENDER2  1.756e-01  1.422e-01  3.221e+02   1.235  0.21776
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) year   age_y  RAGEND yr:g_y
## year        -0.293
## age_y       -0.991  0.259
## RAGENDER2   -0.348  0.140  0.256
## year:age_y   0.497 -0.960 -0.479 -0.122
## y:RAGENDER2  0.159 -0.294 -0.094 -0.640  0.187
## optimizer (nloptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.0233733 (tol = 0.002, component 1)
```

7.

Interpret the fitted multilevel model from question 6 in practical terms.

Model:

$$BP_{ij} = \beta_0+\beta_1 YEAR_{ij}+\beta_2 SEX_i+\beta_3 AGE_i+\gamma_{sex} SEX_i\times YEAR_{ij}+\gamma_{age} AGE_i\times YEAR_{ij}+\gamma_{0j}+\gamma_{1j} YEAR_{ij}+e_{ij}$$

Interpretations on the fitted parameters are as follows:

- *Fixed effect intercept = 72.4703*: This is the overall average blood pressure at the start of the study, where all other variables are set to 0. That is, we have no information on gender, year is 0, and age is 0. This last point does not make any sense practically, but this parameter must be included to preserve model fit (it is very statistically significant, justifying its inclusion). The parameter does not have a proper interpretation
- *Fixed effect of year = 0.7067*: This is the overall effect of year, the positive value suggesting that baseline blood pressure is expected to increase *longitudinally*. This gives an answer to the first study objective. Notably, this parameter is *not* significant, likely because of its redundancy with age.
- *Fixed effect of age = 0.9421*: As participants age, their blood pressure is expected to increase by this amount every year, matching our intuition from the scatter plot in the data exploration. This parameter is extremely significant, meaning age has a strong effect on *initial* blood pressure.
- *Fixed effect of gender = -5.2778*: Since females are encoded with 2 and males with 1, a negative value here means that women have lower systolic blood pressure than men. This parameter is significant, meaning gender is a strong indicator of *initial* blood pressure.
- *Fixed effect of interaction between year and age = -0.0218*: This parameter, along with the next one, both help the researcher answer the second research objective. It is somewhat difficult to interpret in this specification, but the significance of the parameter means that *age has a significant effect on the trajectories of blood pressure over time.*

- *Fixed effect of interaction between year and gender = 0.1756*: Since this parameter estimate is NOT significant, we can conclude that *sex does not have a significant effect on the trajectories of blood pressure over time*, despite it having a significant effect on starting blood pressure.
- *Variance in random intercepts = 169.2586*: There seems to be a fair amount of variance in the random intercepts, suggesting different subjects cover a wide range of initial blood pressure.
- *Variance in random slopes of year = 0.4838*: There is some variance in the trajectories, again reaffirming that the change in blood pressure over time is different across patients. Notably, this value is lower than the initial model, suggesting that the inclusion of sex and age accounts for some of the original variability in blood pressure trajectories.
- *Correlation between random slopes and random intercepts = -0.54*: This has a similar interpretation as before - lower intercepts (i.e. lower baseline blood pressures) are generally expected to have greater slopes (i.e. their blood pressure would increase over time at a higher rate than those with higher initial blood pressure).
- *Variance in residuals = 161.9762*: The leftover is slightly lower than the initial model, meaning this model has captured a bit more of the noise in the data. This could be indicative of a (slightly) improved model fit.

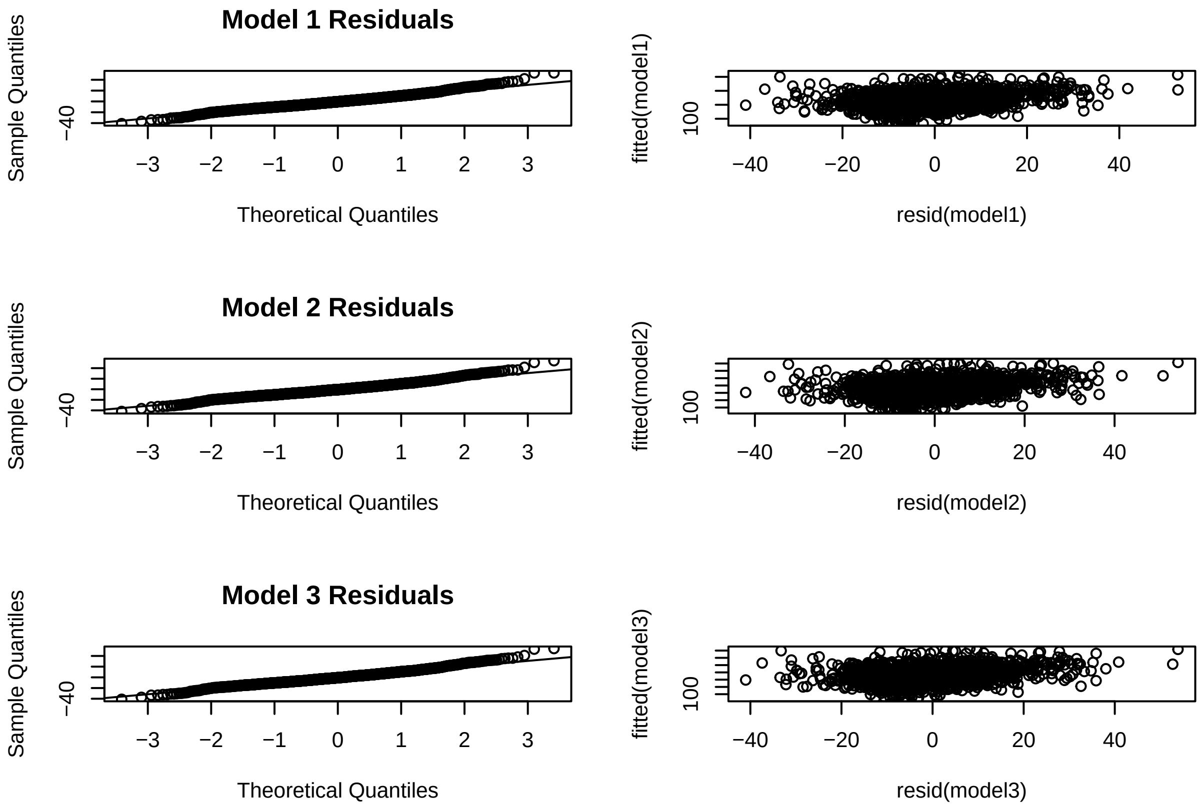
8.

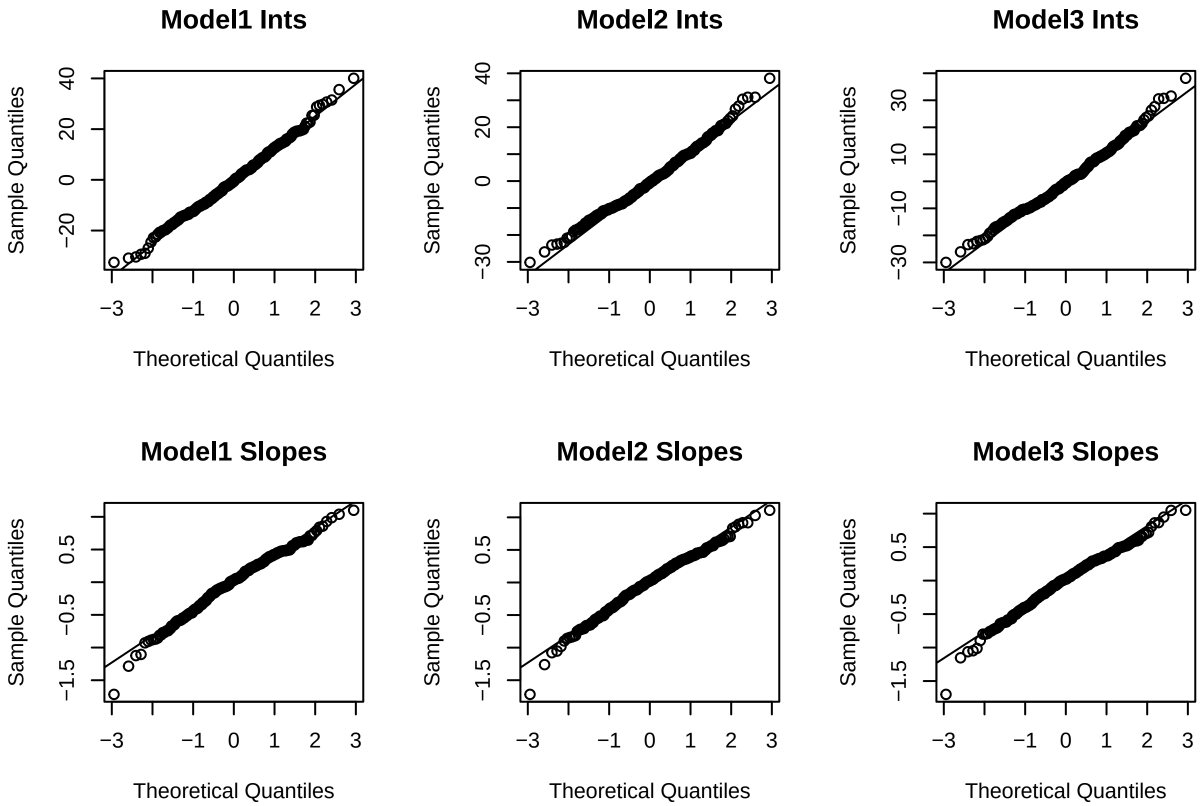
How might one recode the age variable to make the parameters more interpretable? How would this recode change the interpretation of the parameters from 7? Should age and year both be included in the model (be sure to describe your reasoning)?

Encoding age as a categorical variable might be beneficial because then it would clear up its collinearity with year, to an extent. Since both are continuous and are highly correlated, the interpretation of the previous model is somewhat challenging. However, a categorical variable (perhaps one that divides the variable into quartiles, 0 for the youngest quartile, up to 3 for the oldest), would help the interpretation. As it relates to the previous problem, we would be able to determine if specific age groups have more variability in trajectories than others. Intuition may suggest that older patients would have less variability in their blood pressure, because it may have "evened out" in their old age.
Alternatively, one might want age to be a *fixed* value for each participant, since the effect of time is already captured by the year variable. The most logical choice would be the age of the patient when they entered the study, which simplifies interpretation because we would only have to worry about how their *baseline* age effects systolic blood pressure. When we add year as a random effect, the age variable in its current state essentially becomes redundant. This could be done in combination with the quartiles idea, or not.
Age and year likely should both be included, because they represent different things. In the model, year represents the change in blood pressure over time, whereas age represents the actual affect of how old a patient is has on their blood pressure. These are similar, but an important distinction must be made.

9.

Which model (question 2 or 6) appears to have a better fit? Make use of some simple model diagnostics to help justify your choice.


Model 1 Residuals
Sample Quantiles
−40
−3
−2
−1
0
1
2
3
Theoretical Quantiles
fitted(model1)
100
−40
−20
0
20
40
resid(model1)
Model 2 Residuals
Sample Quantiles
−40
−3
−2
−1
0
1
2
3
Theoretical Quantiles
fitted(model2)
100
−40
−20
0
20
40
resid(model2)
Model 3 Residuals
Sample Quantiles
−40
−3
−2
−1
0
1
2
3
Theoretical Quantiles
fitted(model3)
100
−40
−20
0
20
40
resid(model3)




```
##            Base.Model Age...Sex.Included Interaction.Mod
## AIC        12986.4723         12951.0649      12956.4248
## BIC        13018.5677         12993.8588      13009.9171
## ResidVar     162.3796           162.1771        161.9762
```

All models appear to have solid residual diagnostics, with very similar qq-plots. Each have slightly heavy tails, but any differences between them are negligible. All models also have random effects (slopes and intercepts) that are normally distributed, demonstrated by the second set of qq-plots.

The second model, with age and sex included, has the lowest AIC and BIC values along with the lowest residual variance, so it may be preferred as the model with the best fit. However, model three does more to answer the second research question, so it may be preferred. I would choose model 3 for these reasons: 1) an improved fit (regardless of the amount of improvement) compared to the basic model provides a better understanding of the data, 2) the model incorporates all variables and interactions that must be used to explore the research objectives, and 3) the model does not rely on being overly complex to achieve a better fit - it is still interpretable, and the criterion which penalize for the number of parameters being estimated do not penalize the model enough for it to be deemed a worse fit.

Though the diagnostics are similar across each model, which may not suggest a significant improvement in fit between them but at least reaffirms that our models with more parameters of interest are not performing *worse*, boosting their value in answering the research questions. It is quite easy to debate that model 2 ought to be chosen, another strong reason being the warnings associated with convergence and singularity of the third model.

10.

Write a half-page summary of the methods and results of the above analyses regarding the study objectives.

Researchers were interested in how much systolic blood pressure changes over time as people aged, and to quantify the variance in trajectories of blood pressure over time while determining the effect of a patients' sex and age on said trajectories. The first step was exploratory data analysis, which demonstrated a relationship between age and blood pressure, and that there were many different trajectories among subjects. The effect of age and gender on these trajectories seemed difficult to discern, if not minimal. Exploration also revealed a discrepancy between the ages of men and women in the study.

To address the objectives mathematically, multi-level models were fitted consisting of random effects to capture the between-subject variability. In other words, the slopes and intercepts of blood pressure over time were allowed to vary by patient. The initial model simply established that there is large variability in the slopes and intercepts of blood pressure trajectories, and that *blood pressure is generally expected to increase over time*. The second model incorporated sex and age as covariates, the former leading to the conclusion that males have significantly higher baseline blood pressures, the latter that old age also increases baseline blood pressure. Variance in random effects remained fairly consistent. However, the sign of the fixed effect for year became negative, contradicting the conclusion from the first model. The motivation for a third model was to answer the second research objective more directly and to potentially provide more intuitive parameter estimates. Model 3 included interaction terms with time for both sex and age, which allowed for inference on the second objective. It was found that the interaction term between age and time was significant, but the interaction between sex and time was not. *This suggests that age has a significant effect on the trajectories of blood pressure over time, but sex does not.* All three models had very similar diagnostics with normally distributed residuals, normally distributed random effects, and comparable AICs, BICS, and total residual variance. However, Model 2 or Model 3 would be selected as the best model - 2 for the best diagnostics, and 3 for its utility in addressing both research objectives most effectively.

Many additional models were fit and tested, including but not limited to: 1) alternative versions of each model which did not estimate the covariance between random slopes and intercepts, all of which had significantly worse fits by likelihood ratio tests, 2) sex and age as random effects, which showed minimal evidence in the variance of either effect (especially for age), and 3) interaction terms as random effects, which made interpretation very challenging, worsened model criteria, and faced convergence or singularity issues. The final models included in this report were deemed to be the most useful in demonstrating the work done.

Future work can certainly be done on this subject, primarily as it relates to the recoding of the age variable which is currently redundant of year. While this redundancy may not matter if we were only interested in questions relating to other variables, we were specifically aiming to find the effect of age on variability in blood pressure trajectories. Thus, additional analysis which better handles this variable would be quite valuable. Also, more control on accounting for the discrepancy in age between men and women in the study may also be useful, though this may already be sufficiently captured by the inclusion of both variables in the second and third models.

Appendix

Additional models I tried but ultimately excluded

```
### other modeling attempts
# age and year and naturally very highly correlated, they both increase monotonically,
```

```r
#   try a model with year only as a random effect, assuming its contribution is
#   captured by age_y
lmer(bp_sys ~ age_y + factor(RAGENDER) + (year|HHIDPN),
               REML=T,data=hrs_wbbp_long) %>% summary() # failed to converge

# include interaction, based on boxplot from EDA which showed differences
#   in starting age for males vs females
lmer(bp_sys ~ age_y + factor(RAGENDER) + age_y:factor(RAGENDER) + (year|HHIDPN),
               REML=T,data=hrs_wbbp_long) %>% summary()
# still very poor AIC, w/ and w/o year as its own fixed effect. w/ fails to converge


lmer(bp_sys ~ year + age_y + RAGENDER +
       (year||HHIDPN) + (age_y||HHIDPN),
     REML = T,data=hrs_wbbp_long) %>% summary()
# Worse AIC and singular fit, age random effect has 0 variance regardless of
# whether it is included as a fixed effect as well. since this model assumes
# the effect of age to vary across participants, but finds no variance, we can
# probably rule this model out (along with the AIC and singularity reasons). We
# can also probably rule out any models that use age as a random effect.

# intercepts-only interaction model - similar AIC
lmer(bp_sys ~ year + age_y + RAGENDER + year:age_y + year:RAGENDER +
       (1|HHIDPN),
     REML = T,data=hrs_wbbp_long) %>% AIC()
```